

Data Visualization Analysis Report: Sample Superstore Dataset

Data Analysis Submission
April 3, 2026

Introduction

This report presents the required visualizations for the Sample Superstore dataset. Each chart is placed in its proper section with a short interpretation to explain the business insight drawn from the visual.

1. Line Chart: Sales Over Time (2014–2017)

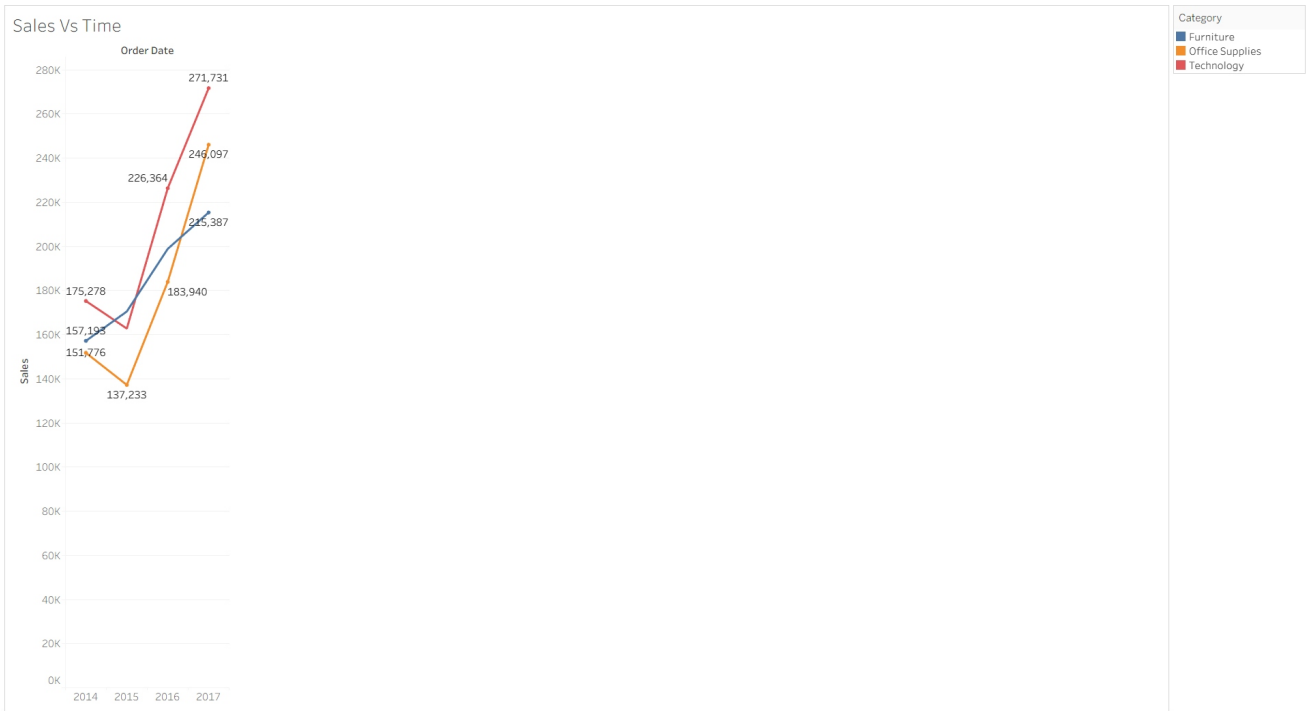


Figure 1. Sales vs Time Line Chart

Interpretation: This line chart shows category-wise sales from 2014 to 2017. Technology records the highest sales by 2017, while Office Supplies also shows strong growth. The pattern suggests that sales accelerated after 2015, especially in the final year.

2. Scatter Plot: Relationship Between Sales and Profit

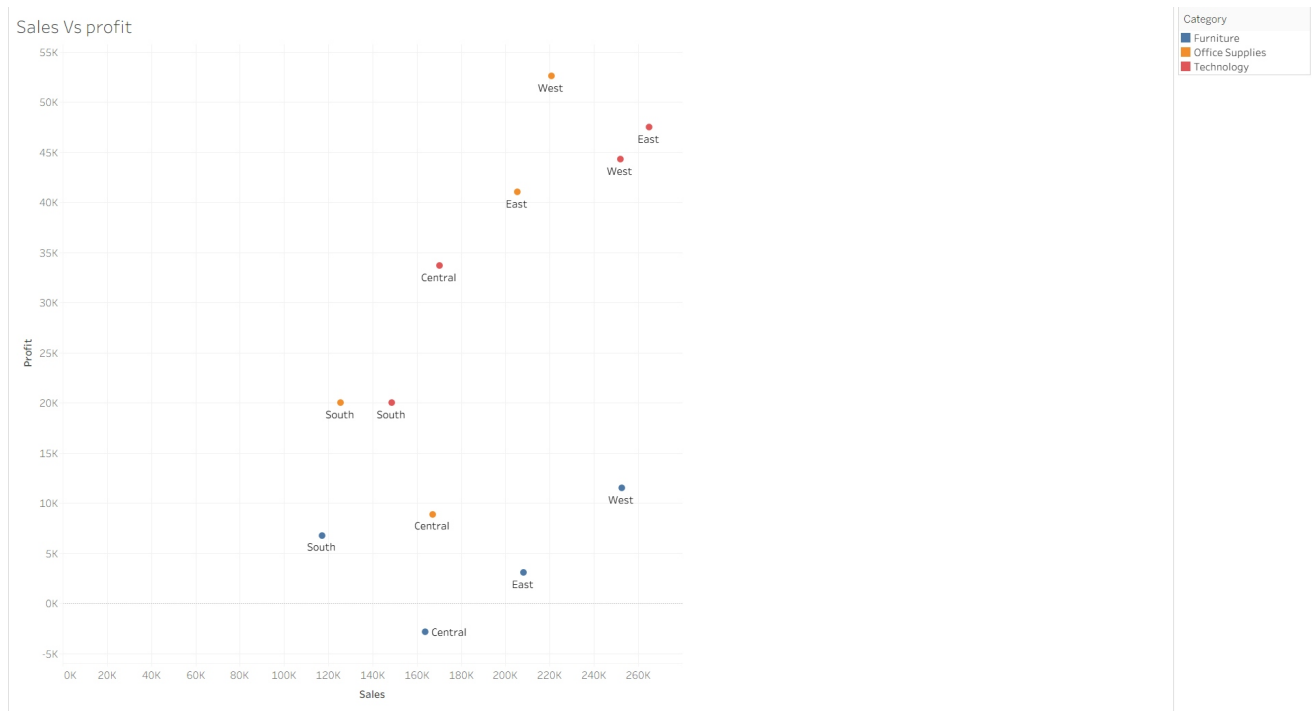


Figure 2. Sales vs Profit Scatter Plot

Interpretation: The scatter plot indicates a generally positive relationship between sales and profit, but it also highlights outliers. Furniture in the Central region appears as a loss-making point despite decent sales, while Technology performs better in East and West regions.

3. Bar Chart: Category and Sub-category Analysis

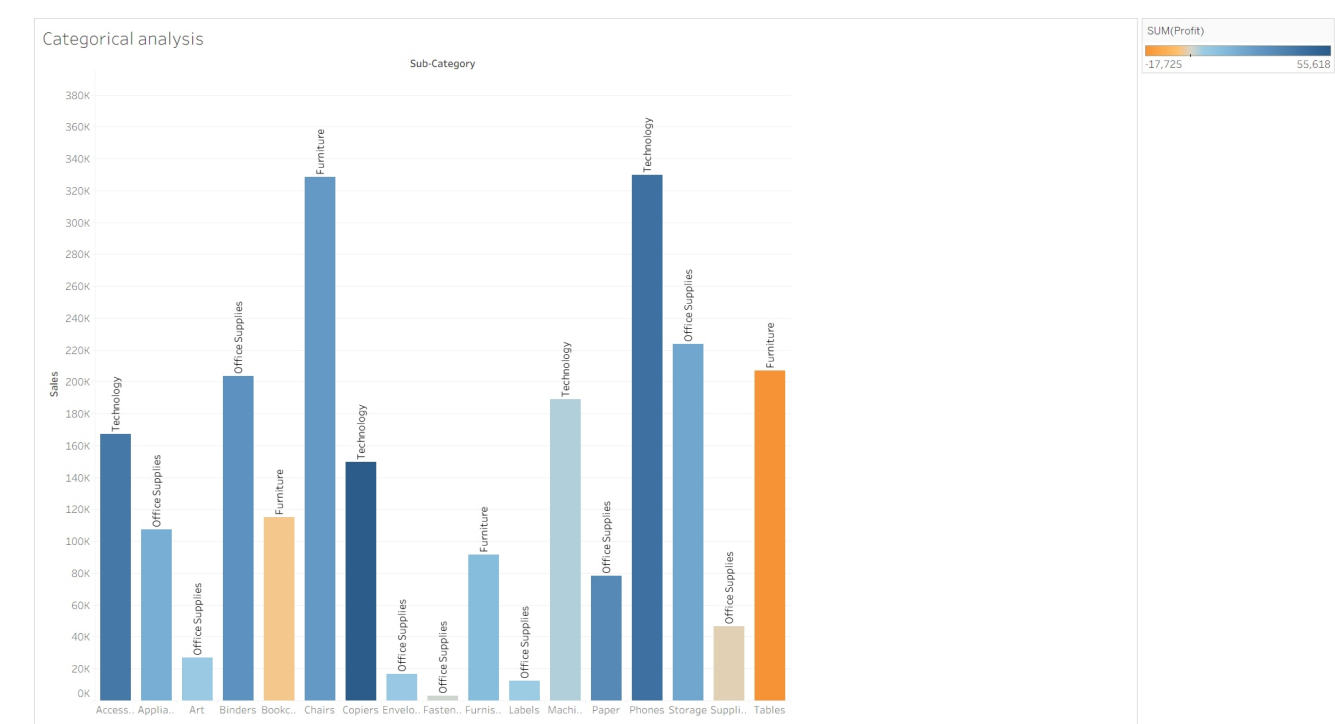


Figure 3. Category and Sub-category Bar Chart

Interpretation: The bar chart compares sub-categories by sales and profit contribution. Phones and Chairs generate strong sales, while Tables and some furniture-related items appear weaker in profit performance. This helps identify top-performing and loss-making areas.

4. Pie Chart: Region-wise Contribution

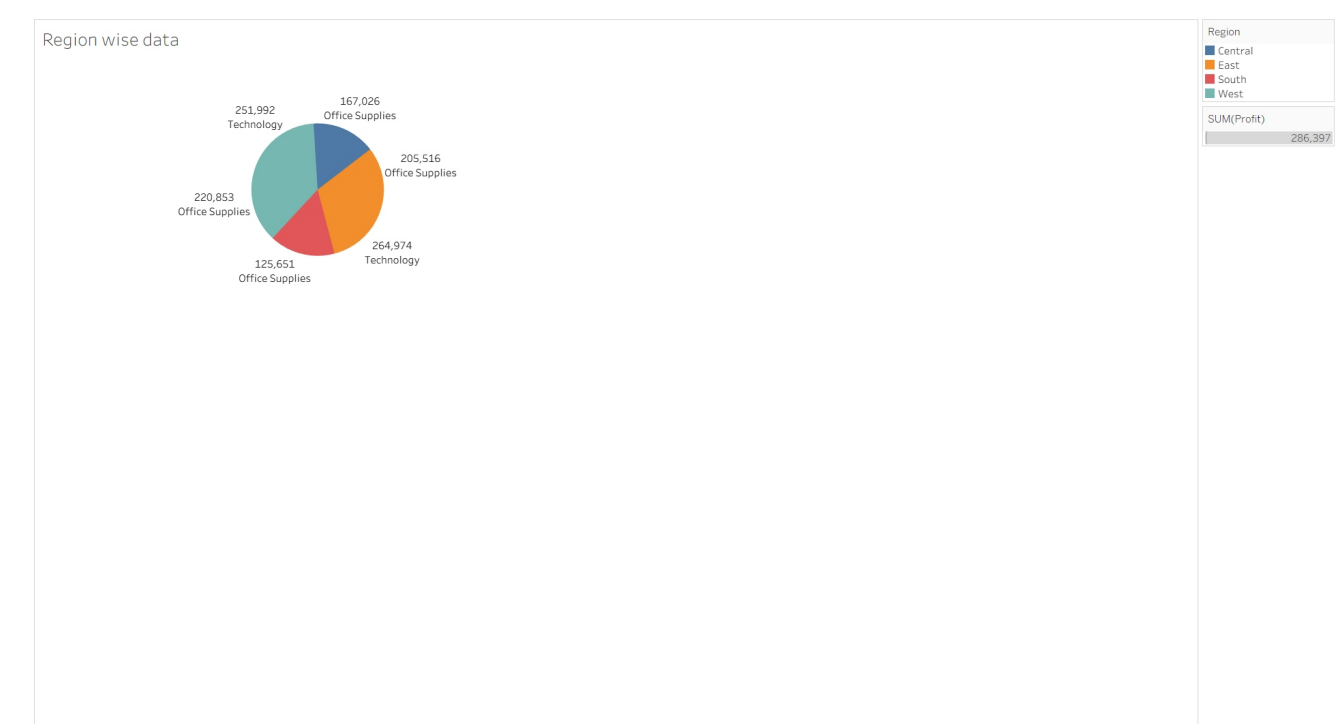


Figure 4. Region-wise Pie Chart

Interpretation: The pie chart summarizes the regional contribution to overall business performance. West and East contribute the largest shares, whereas South contributes less, which may point to lower penetration or weaker demand in that region.

5. Trend Line: Discount Patterns

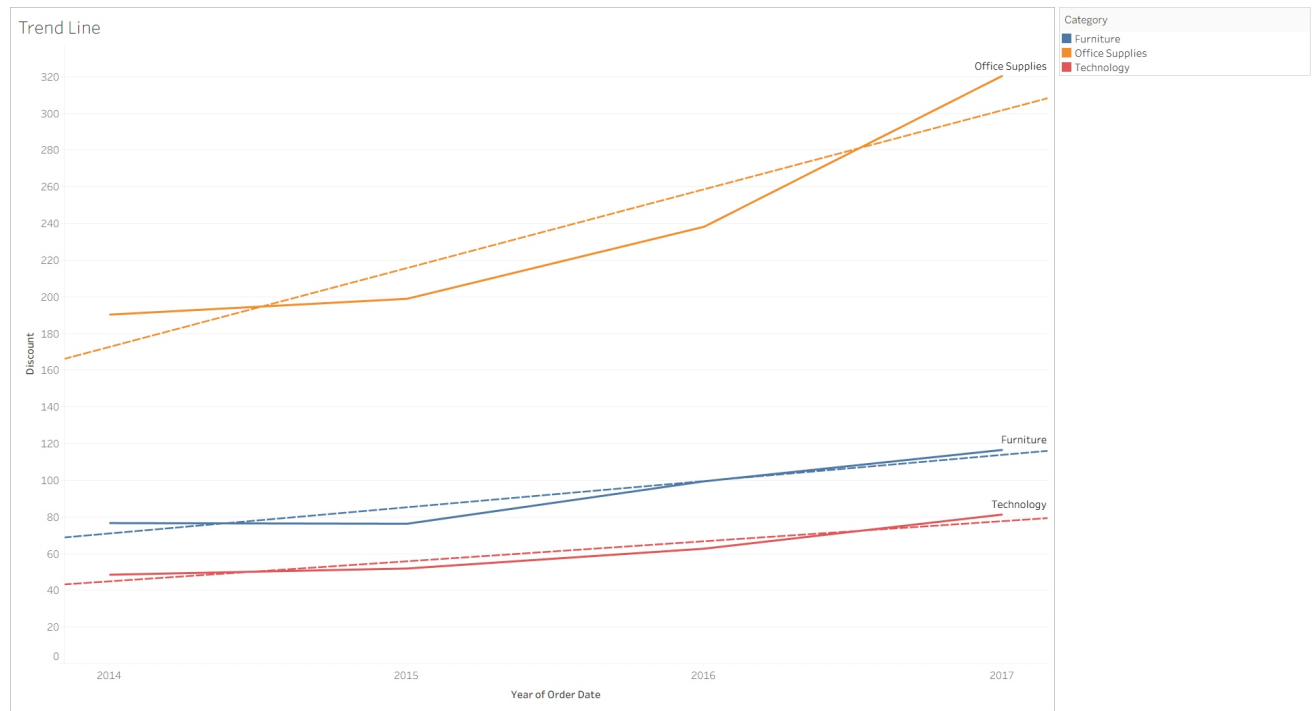


Figure 5. Discount Trend Line

Interpretation: The trend line shows that discounts have increased over time across categories, especially for Office Supplies. This may support sales growth, but rising discount levels can reduce profit margins if not controlled properly.

6. Histogram: Distribution of Profit

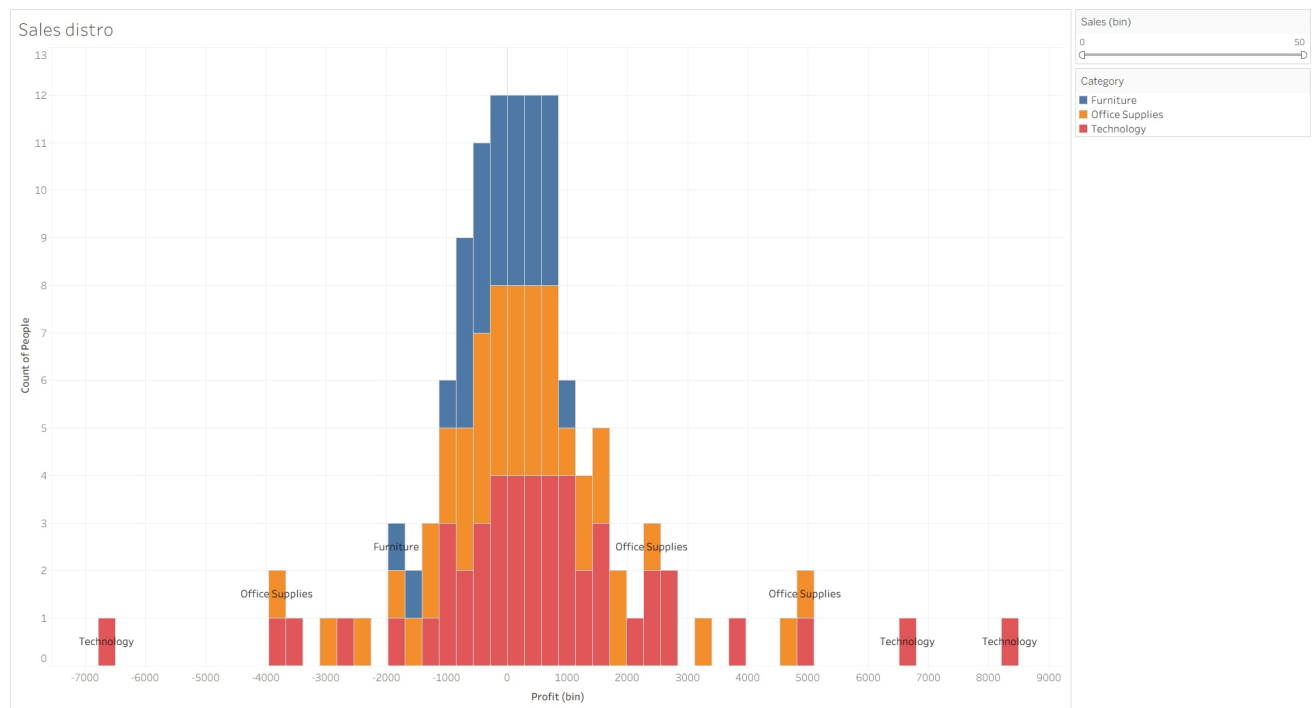


Figure 6. Profit Distribution Histogram

Interpretation: The histogram shows that most profit values are concentrated near zero, meaning many transactions generate only small gains or losses. A few extreme negative and positive values suggest volatility in profitability across orders.

